

Sequencing

... For library preparation we used a rapid barcoding kit (RBK114.24 V14). We then sequence using an ONT Minion with a R9.4 flow cell.

assembly

For assembly, we merged the separate barcodes for the *B. canis* sample together and then ran the merged fastq file through Raven (citation) version 1.8.3. We then used blast (reference) to identify which contig belonged to which *B. canis* chromosome.

Next, we reduced the number of errors in our contigs by polishing and an indel clean up step. For polishing we used minimap2 (citations) version 2.28-r1209 with the -a to map reads to the assembly. Then, we polished the assembly with tbCon version 2024-08-07 from bioTools <https://github.com/jeremyButtler/bioTools> using a percent SNP support of 30% (-perc-snp-sup 0.3), a minimum read depth of 5 (-min-depth 5), a minimum base q-score of 3 (-min-q 3), and a minimum insertion q-score of 3 (-min-q-ins 3). We polished the first chromosome twice and the second chromosome once.

For indel clean up we mapped the assembly to the *B. canis* CP016977 and CP016978 reference genomes using minimap2 with the -a flag. We then removed all indels less than four bases using rmHomo from bioTools version 2025-07-28 with the -homo 1 flag.

cgMLST tree

We then built a cgMLST reference database for our in house cgMLST scripts using the Rimsphere database *B. canis* (citation). Then we used inhouse scripts to find the cgMLST lineages for the 31 full length reference genomes we found on Genbank (Supplemental 1, Eric this is in the 01-input/01-references.acc file) and our cleaned assembly. We then built a minimum spanning cgMLST tree with our database using GrapeTree (citation) with the -matrix MSTreeV2 flag. The final tree was edited and saved using Phylo.io <https://beta.phylo.io/viewer>.

For the genomic tree, we extracted all cgMLST genes from the references, assembly, and our outgroup (CP000911; a *B. susis* genome) using in house scripts. We then ... (Eric this is your part if you took this further).

To see our in house scripts or more detailed step by step instructions please visit <https://github.com/jeremyButt/2024-UKR-Bcanis>.